

Labour Supply – Individual Attachment to the Labour Market

ECON 300 - Labour Economics

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Learning Objectives

- Define employment, unemployment, labour force participation, and hours worked; understand Statistics Canada measurement and reporting.
- Use the income–leisure model to show trade-offs in whether and how much to work.
- Distinguish individual choices from the opportunity sets available.
- Decompose wage changes into income and substitution effects and interpret ambiguous net responses.
- Apply the model to married women's labour supply decisions.

Main Questions

- How do we measure labour market attachment, and how has it evolved?
- Where does participation fit within labour supply theory?
- How do men's and women's labour supply decisions differ and why?
- Should labour supply increase with the market wage? What does theory say?
- What does the evidence show about wage responsiveness?
- Why do firms pay an *overtime premium* instead of just a higher straight-time wage?

Quantifying Labour Market Attachment

Labour Force (LF): Eligible population (15+) who are employed or unemployed.

LF Participation Rate (LFPR): Fraction of eligible population in LF; $LFPR = \frac{LF}{POP}$.

Student Tip

Keep POP (eligible), LF (in the game), and those *outside* LF distinct; misclassification drives measurement issues.

Unemployment (Definition)

To be **unemployed**, a person must be in one of:

1. Without work and actively searched in the last 4 weeks,
2. Waiting to be recalled from layoff,
3. Waiting to start a job within 4 weeks.

Figure 2.1: Labour Force Concepts (Jan 2020)

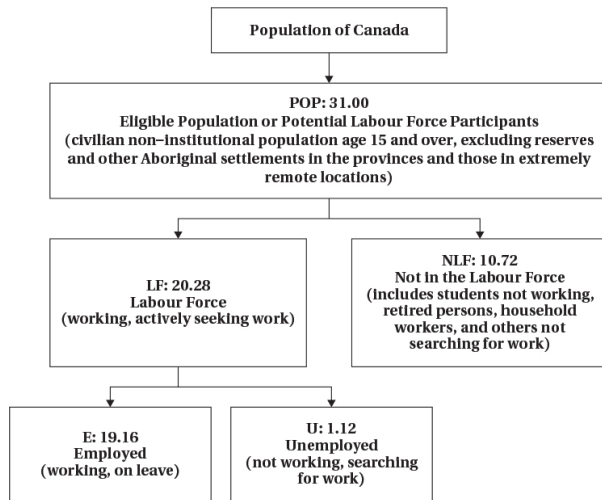


Figure 2.2: LF Participation Rates by Sex, Canada, 1901–2019

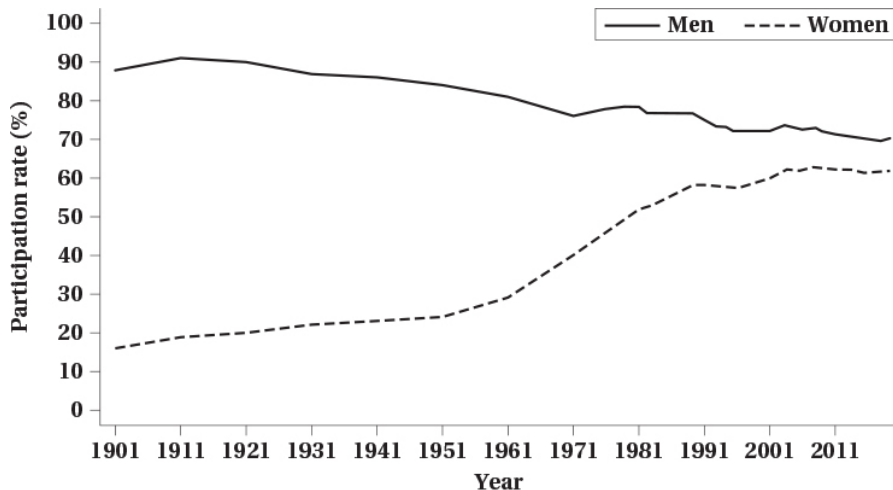
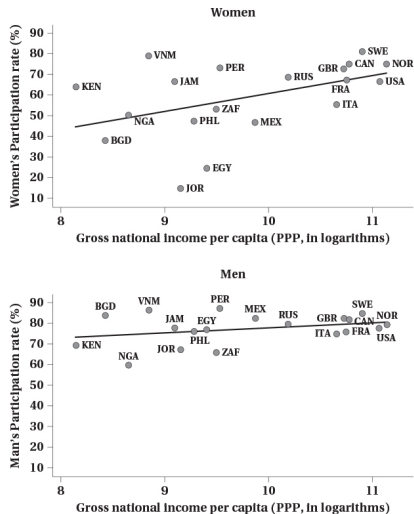


TABLE 2.1: Labour Force Participation Rates by Sex, Selected Countries, 2018

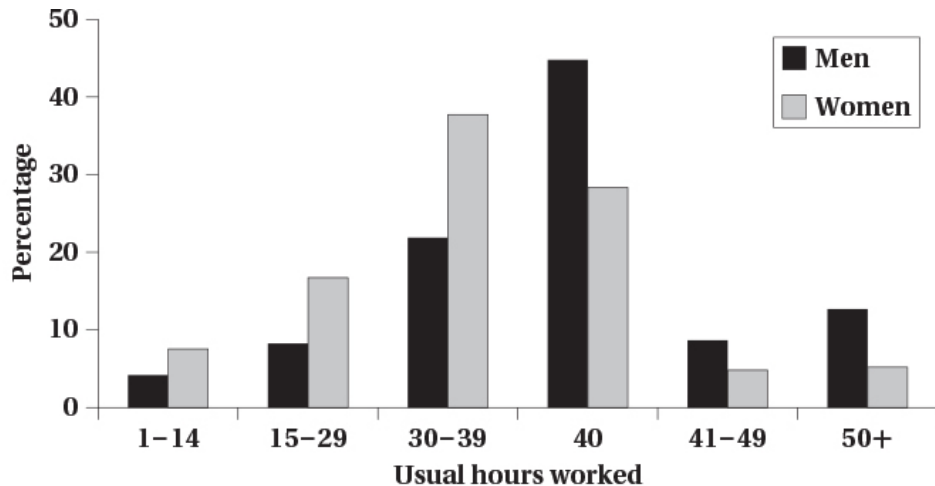
Country	Abbrev.	Overall	Male	Female
Sweden	SWE	82.9	84.8	81.0
Vietnam	VNM	82.7	86.4	79.1
Peru	PER	80.3	887.2	73.3
Canada	CAN	78.5	81.9	75.1
United Kingdom	GBR	77.5	82.2	72.8
Norway	NOR	77.3	79.2	75.2
Russia	RUS	74.1	79.8	68.9
Jamaica	JAM	72.4	78.0	66.9
United States	USA	72.3	77.7	66.8
France	FRA	71.7	75.8	67.5
Kenya	KEN	66.9	69.6	64.1
Italy	ITA	65.3	74.9	55.7
Mexico	MEX	64.7	82.6	47.1
Philippines	PHL	62.0	76.2	47.7
Bangladesh	BGD	61.1	83.9	38.1
South Africa	ZAF	59.6	66.0	53.4
Nigeria	NGA	55.2	59.9	50.5
Egypt	EGY	51.1	76.9	24.7
Jordan	JOR	41.6	67.4	15.1

Figure 2.3: Male and Female Participation vs. Development



- Hours per day, days per week, weeks per year.
- Affects the *quantity* and *quality* of labour supply.

Figure 2.4: Distribution of Hours Worked Per Week by Sex, 2019



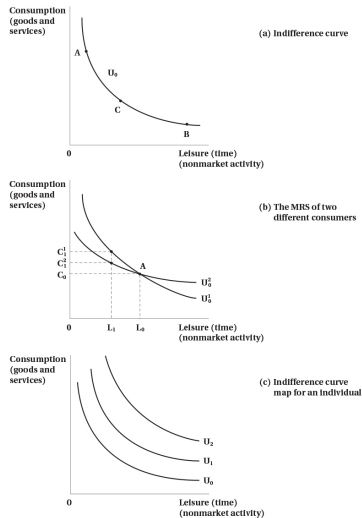
Choice of hours worked given opportunities and value of non-market time:

- **Preferences & Constraints**
- Individuals choose the feasible outcome that maximizes satisfaction.

Two goods: **Consumption** and **Leisure**; represented by indifference curves.

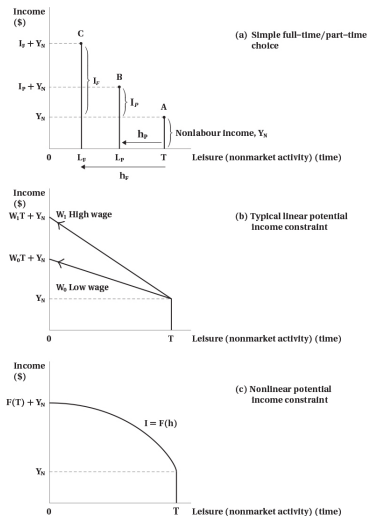
- Well-defined preferences imply every bundle lies on some indifference curve.
- Indifference curves show combinations yielding equal utility.
- $|\text{slope}| = \text{MRS}$: leisure accepted per unit of consumption foregone.

Figure 2.5: Consumption–Leisure Indifference Curves



- Market sets price of consumption; transforms consumption–leisure into **income–leisure**.
- Potential income (budget / full-income) constraint defines feasible set of income–leisure combinations.

Figure 2.6: Potential Income Constraints

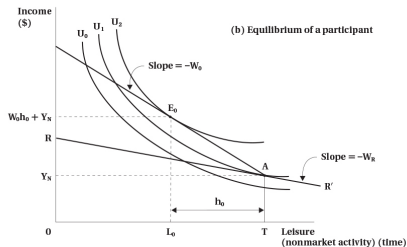
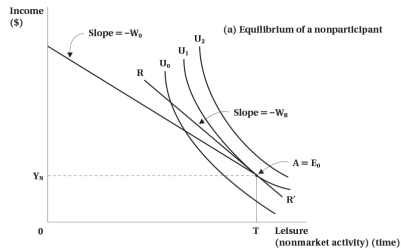


The Consumer's Optimum

Optimal income and leisure at the highest indifference curve tangent to the income constraint.

- Compare **MRS** (willingness to trade leisure for income) with the **market wage** (ability to trade).

Figure 2.7: The Consumer's Labour Supply Decision



Worked Example: Part-Time Work at University

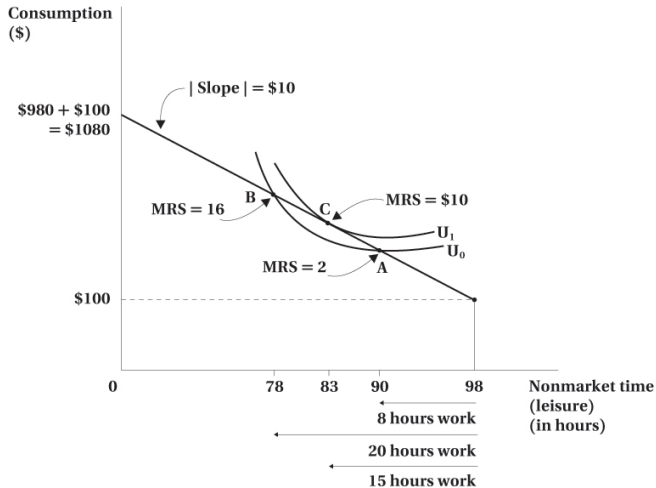
Jack and Jill each have a weekly allowance of \$100 and can work at \$10/hour. They have 98 discretionary hours (after 10 hours/day personal needs).

Budget line: $C = 100 + 10H$, where H is hours worked; leisure $L = 98 - H$.

Interpretation: Intercepts at $C = 100$ (no work) and $H = 98$ hours $\rightarrow C = 100 + 10(98) = 1080$.

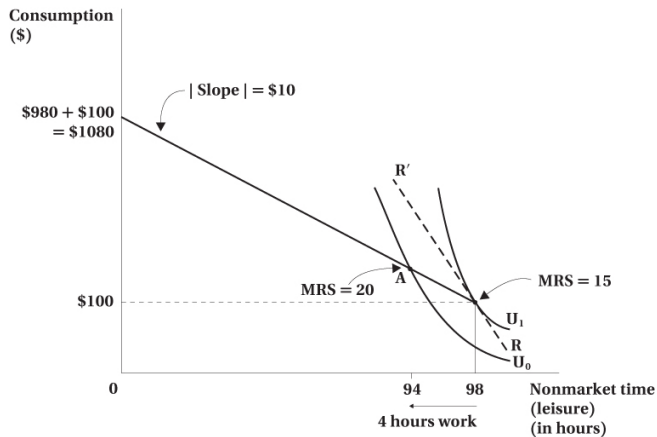
Use case: Place indifference curves to show who works more given identical opportunities but different preferences for leisure vs. consumption.

Worked Example: Part-Time Work at University



(a) Jack

Worked Example: Part-Time Work at University



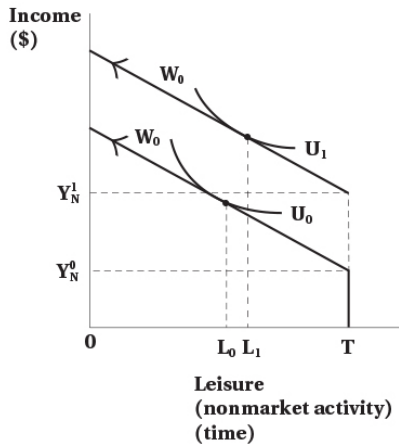
(b) Jill

Increase in non-labour income shifts the budget constraint outward (parallel).

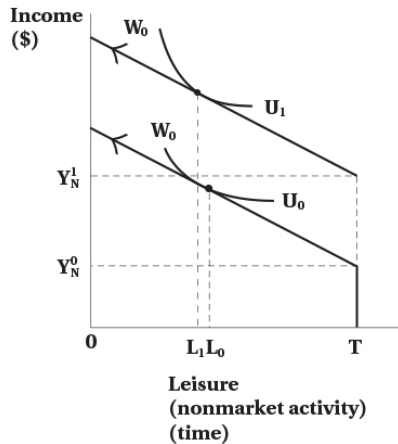
- If leisure is **normal**, consume more leisure $\Rightarrow$ fewer hours.
- If leisure is **inferior**, consume less leisure $\Rightarrow$ more hours.

Figure 2.8: Increase in Non-Labour Income

(a) Leisure a normal good



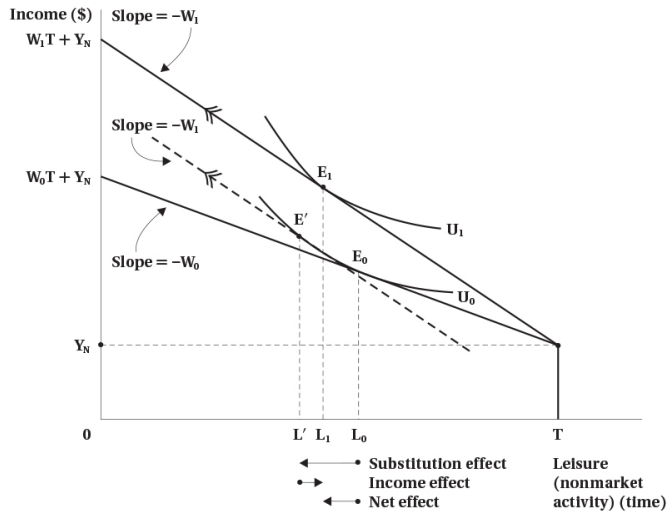
(b) Leisure an inferior good



Wage Changes: Income vs. Substitution Effects

- **Income effect:** Higher wage $\Rightarrow$ higher income $\Rightarrow$ more leisure (fewer hours).
- **Substitution effect:** Higher wage raises price of leisure $\Rightarrow$ substitute toward work (more hours).
- Net effect on hours is **ambiguous**.

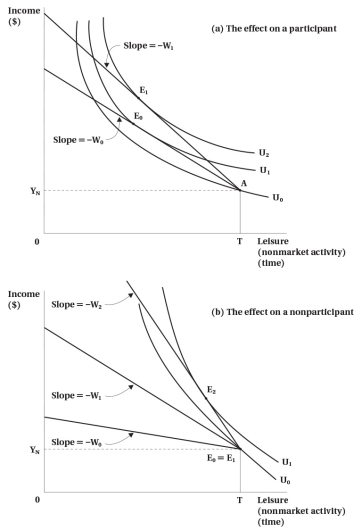
Figure 2.9: Decomposing a Wage Increase



Participation Effects of Wage Changes

- For a non-participant, higher wage can induce entry if it exceeds reservation wage.
- For participants, hours may rise or fall depending on which effect dominates.

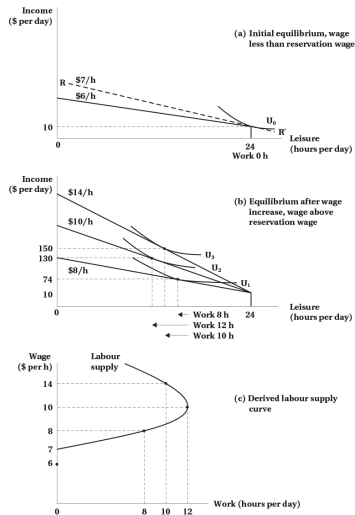
Figure 2.10: Wage Increase and Participation



Deriving the Individual Labour Supply Curve

If substitution effect dominates at low wages, hours increase with wage; at higher wages, income effect can dominate, producing a **backward-bending** segment.

Figure 2.11: Individual Labour Supply Curve



- Attachment has two margins: **participation** and **hours**.
- The consumer choice model captures both: choose income (consumption) and leisure within a potential income constraint.
- **Reservation wage** governs participation: above it $\Rightarrow$ work; below it $\Rightarrow$ no participation.
- Varying the wage traces an individual's labour supply; income vs. substitution effects can bend supply backward.

Thank You!